

Dwarf planets

In 2006 the International Astronomical Union defined a planet as a body that orbits the Sun, is round under its own gravity, and has cleared the neighbourhood around its orbit. A body that meets the first two conditions but not the third is a dwarf planet. Pluto, which shares its region with many Kuiper belt objects, was reclassified under this definition.

Five bodies are officially recognised as dwarf planets: Ceres, Pluto, Haumea, Makemake and Eris. Ceres is the largest object in the asteroid belt, about 940 kilometres across, and the only dwarf planet in the inner Solar System. The Dawn spacecraft orbited it from 2015 to 2018 and found bright deposits of salt on its surface.

Pluto has a diameter of about 2,377 kilometres, smaller than Earth's Moon. Its largest moon, Charon, is about half its size, so the two orbit a point in space between them. The New Horizons spacecraft flew past Pluto in July 2015 and revealed a young, heart-shaped plain of nitrogen ice called Sputnik Planitia, mountains of water ice, and a thin nitrogen atmosphere.

Eris, discovered in 2005, is slightly smaller than Pluto but about 27 percent more massive, and its discovery was the immediate trigger for the 2006 definition. Haumea is unusual for its elongated shape, the result of a rotation period of under four hours. Makemake is a reddish body a little smaller than Pluto with one known moon.